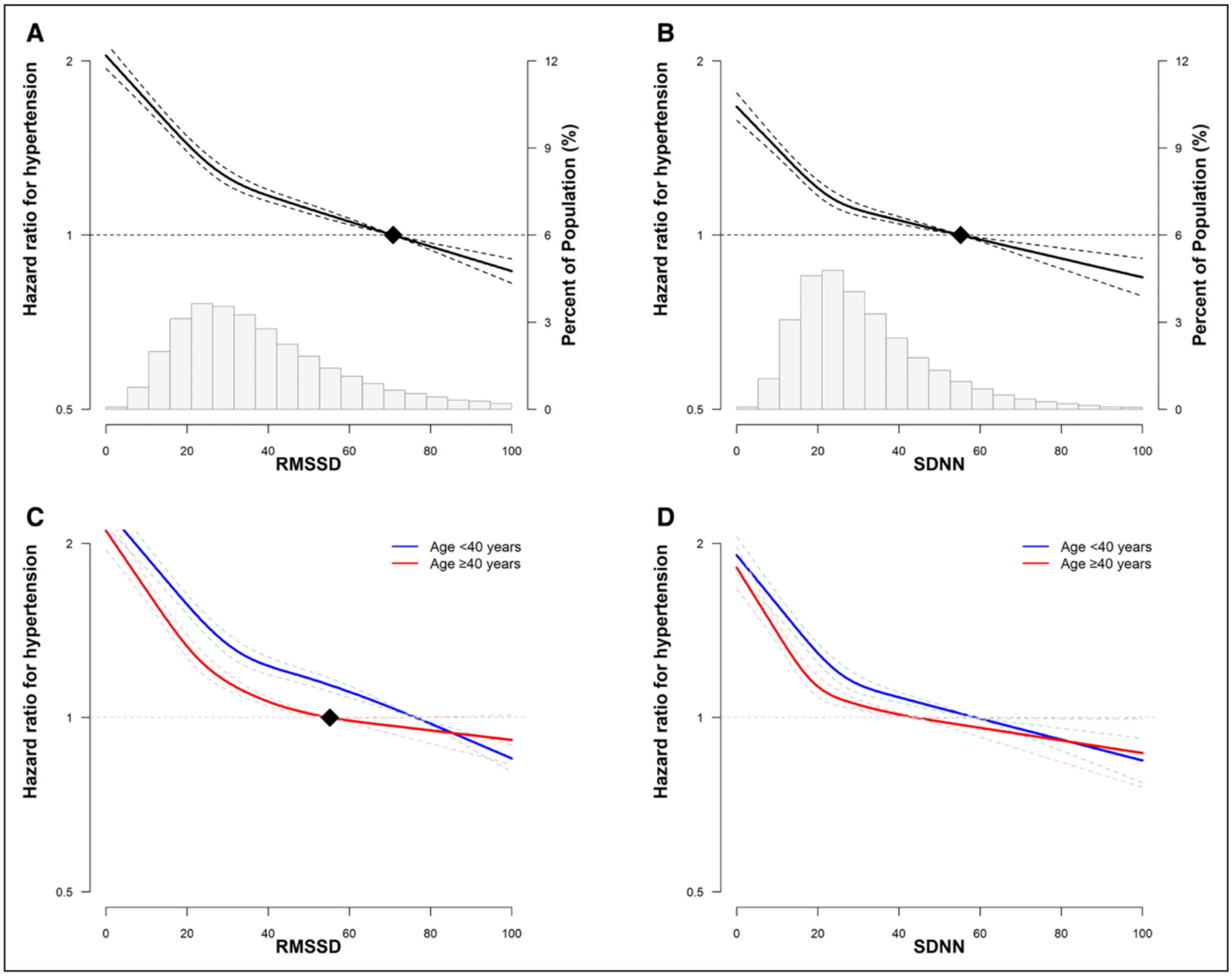

Figure. Multivariable-adjusted hazard ratios for hypertension.
A, Root mean square of successive R-R interval difference (RMSSD) in the total population, (**B**) SD of normal-to-normal R-R interval (SDNN) in the total population, (**C**) RMSSD in participants aged <40 and ≥40 y, and (**D**) SDNN in participants aged <40 and ≥40 y. The curves represent adjusted hazard ratios for hypertension based on restricted cubic splines with knots at the 5th, 35th, 65th, and 95th percentiles of the sample distribution. The models were adjusted for age, sex, center, year of screening examination, alcohol consumption, smoking, physical activity, total energy intake, sleep duration, education level, depression, medication for hyperlipidemia, history of diabetes, and body mass index.

from 10-second, 12-lead electrocardiography was an independent predictor of all-cause mortality in patients with ST-segment–elevation myocardial infarction.[33] Furthermore, HRV on a 10-second electrocardiography was also associated with various cardiovascular conditions, such as cardiovascular mortality, heart failure, and atrial fibrillation, in the general population.[6,34,35] Although an accumulating body of evidence has suggested that 10-second electrocardiography may be sufficient to predict clinically important CVD outcomes, it is unclear whether 10-second electrocardiography predicts hypertension in young, asymptomatic individuals. Our study is the first to show that 10-second HRV recording, as indexed by RMSSD and SDNN, was significantly and inversely associated with the risk of clinical hypertension, highlighting its potential as a simple and easily accessible tool to predict incident hypertension in a clinical setting. In our study, RMSSD showed a slightly stronger association with incident hypertension than SDNN, suggesting that RMSSD may be a better index of autonomic function based on this 10-second electrocardiography recording. The prognostic value of SDNN in 10-second HRV recording is unclear as SDNN is less reliable than other HRV parameters, such as RMSSD, in short-term HRV recording.[36] Previous validation studies have also indicated that a single 10-second electrocardiography recording yields valid RMSSD measurements, while SDNN requires a longer duration of >30 seconds.[37,38] However, our findings demonstrated independent associations between SDNN and RMSSD and hypertension, suggesting that both parameters on ultrashort-term electrocardiography may be useful in predicting hypertension development.

Our study results support the hypothesis that autonomic nervous system dysregulation precedes hypertension.[39] The exact cause underlying the mechanism

Table 4. Development of Hypertension by RMSSD Change (n=150 301)

Quintiles of RMSSD change	PY	Incident cases	Incidence rate (/10³ PY)	Age- and sex-adjusted HR (95% CI)	Multivariable-adjusted HR* (95% CI) Model 1	Model 2	HR (95% CI)† in model using time-dependent variables
Total							
Q1 (<−15.34)	91 447	3560	38.9	0.89 (0.85–0.93)	1.19 (1.14–1.26)	1.16 (1.10–1.22)	1.05 (1.01–1.10)
Q2 (−15.34 to −4.76)	94 127	4335	46.1	0.96 (0.92–1.00)	1.04 (0.99–1.08)	1.04 (1.00–1.09)	1.02 (0.98–1.06)
Q3 (−4.75 to 2.78)	95 745	4775	49.9	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Q4 (2.79 to 12.70)	97 525	4717	48.4	0.99 (0.95–1.03)	0.99 (0.95–1.03)	0.99 (0.95–1.03)	0.91 (0.87–0.94)
Q5 (≥12.71)	97 916	3745	38.2	0.85 (0.82–0.89)	0.89 (0.86–0.93)	0.90 (0.86–0.94)	0.83 (0.79–0.87)
P value for trend				0.255	<0.001	<0.001	<0.001
Age <40 y							
Q1 (<−15.34)	68 302	2295	33.6	0.86 (0.81–0.91)	1.19 (1.11–1.26)	1.15 (1.08–1.23)	1.10 (1.03–1.18)
Q2 (−15.34 to −4.76)	60 885	2345	38.5	0.96 (0.91–1.02)	1.04 (0.99–1.11)	1.05 (0.99–1.11)	1.07 (0.99–1.14)
Q3 (−4.75 to 2.78)	56 739	2281	40.2	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Q4 (2.79 to 12.70)	58 652	2229	38.0	0.96 (0.91–1.02)	0.96 (0.91–1.02)	0.96 (0.90–1.01)	0.94 (0.87–1.01)
Q5 (≥12.71)	66 055	2064	31.2	0.82 (0.77–0.87)	0.86 (0.81–0.91)	0.86 (0.81–0.91)	0.84 (0.77–0.90)
P value for trend				0.242	<0.001	<0.001	<0.001
Age ≥40 y							
Q1 (<−15.34)	23 145	1265	54.7	0.89 (0.83–0.95)	1.21 (1.13–1.30)	1.19 (1.11–1.28)	1.01 (0.96–1.07)
Q2 (−15.34 to −4.76)	33 242	1990	59.9	0.93 (0.88–0.99)	1.02 (0.96–1.08)	1.03 (0.97–1.10)	0.99 (0.94–1.04)
Q3 (−4.75 to 2.78)	39 006	2494	63.9	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Q4 (2.79 to 12.70)	38 873	2488	64.0	1.00 (0.95–1.06)	1.00 (0.95–1.06)	1.00 (0.95–1.06)	0.89 (0.84–0.93)
Q5 (≥12.71)	31 861	1681	52.8	0.85 (0.80–0.91)	0.91 (0.86–0.97)	0.92 (0.87–0.98)	0.81 (0.77–0.86)
P value for trend				0.868	<0.001	<0.001	<0.001

P=0.307 for the overall interaction between age and RMSSD change for hypertension (multivariable-adjusted Model 1). HR indicates hazard ratio; PY, person-years; and RMSSD, root mean square of successive differences in R-R intervals.

*Estimated from Cox proportional hazards models. Model 1 was adjusted for age, sex, center, year of screening examination, alcohol consumption, smoking, physical activity, body mass index, total energy intake, sleep duration, education level, depression, medication for hyperlipidemia, history of diabetes, and RMSSD. Model 2 was adjusted for variables included in Model 1 and heart rate.

†Estimated from Cox proportional hazard models with quintiles of RMSSD change, smoking, alcohol consumption, physical activity, total energy intake, body mass index, sleep duration, depression, medication for hyperlipidemia, and history of diabetes as time-dependent categorical variables, and age, sex, center, year of screening examination, education level, and RMSSD at baseline as time-fixed variables.

by which autonomic dysfunction leads to the onset of hypertension is largely unknown; however, excessive sympathetic drive has been thought to be a prevalent feature in the pathway.[28] Sympathetic overdrive is related to many components of metabolic syndrome[28] and is likely to be present in individuals with hypertension.[40] However, increasing evidence suggests that parasympathetic modulation is also responsible for the development of hypertension.[39] Epidemiological studies have suggested that vagal tone, as measured by HRV, is lower in individuals with hypertension than in those without hypertension, even after adjustment for a range of covariates.[26,28,31] In other studies, increased sympathetic nerves and decreased cardiac vagal activation were observed in the young- or early-onset hypertension, in turn leading to increased SBP, cardiac output, and resting tachycardia.[40–42] In our study, both SDNN, a marker of overall autonomic modulation, and RMSSD, a marker of parasympathetic tone, showed a significant association with incident hypertension,[43] further demonstrating that excessive sympathetic activity and vagal modulation contributes to the development of hypertension. Additional studies are needed to confirm our findings and better elucidate the relative contribution of parasympathetic withdrawal versus sympathetic overactivity in the association between HRV and hypertension.

Our study has some limitations. We measured HRV using a 10-second electrocardiography. However, studies have assessed the reliability and efficacy of 10-second HRV compared with those of 5-minute HRV in the general population and in patients with hypertension.[44,45] Furthermore, we only used time-domain parameters of HRV and not frequency-domain HRV parameters, as 10-second electrocardiography allows estimation of only limited time-domain parameters for HRV. Additionally, BP was determined according to single-day measurements, as in most studies. Although 3 readings were performed, and the average of the second and third readings was used for analysis, a misclassification of BP categories could not be excluded; however, this would have resulted in the underestimation of the true associations. Third, information on diabetic autonomic neuropathy was not available in our study, although

Table 5. Development of Hypertension by SDNN Change (n=150 301)

Quintiles of SDNN change	PY	Incident cases	Incidence rate (/10³ PY)	Age- and sex-adjusted HR (95% CI)	Multivariable-adjusted HR* (95% CI)		HR (95% CI)† in model using time-dependent variables
					Model 1	Model 2	
Total							
Q1 (<−12.84)	92 666	3849	41.5	0.93 (0.89–0.98)	1.17 (1.12–1.24)	1.11 (1.06–1.17)	1.06 (1.01–1.10)
Q2 (−12.84 to −4.12)	94 119	4310	45.8	1.00 (0.96–1.04)	1.07 (1.02–1.11)	1.07 (1.02–1.11)	1.02 (0.98–1.06)
Q3 (−4.11 to 2.26)	95 626	4500	47.1	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Q4 (2.27 to 10.53)	97 281	4435	45.6	0.99 (0.95–1.03)	0.99 (0.95–1.03)	0.98 (0.94–1.03)	0.93 (0.89–0.97)
Q5 (≥10.54)	97 067	4038	41.6	0.91 (0.87–0.95)	0.95 (0.91–0.99)	0.95 (0.91–0.99)	0.95 (0.91–0.99)
P value for trend				0.191	<0.001	<0.001	<0.001
Age <40 y							
Q1 (<−12.84)	67 617	2405	35.6	0.92 (0.87–0.98)	1.20 (1.12–1.28)	1.14 (1.07–1.22)	1.09 (1.02–1.17)
Q2 (−12.84 to −4.12)	60 514	2249	37.2	1.00 (0.94–1.06)	1.07 (1.01–1.13)	1.07 (1.00–1.13)	1.01 (0.94–1.08)
Q3 (−4.11 to 2.26)	57 468	2142	37.3	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Q4 (2.27 to 10.53)	59 592	2204	37.0	1.00 (0.94–1.06)	1.00 (0.94–1.06)	0.99 (0.94–1.06)	0.94 (0.87–1.02)
Q5 (≥10.54)	65 442	2214	33.8	0.90 (0.84–0.95)	0.92 (0.87–0.98)	0.92 (0.87–0.98)	0.92 (0.85–0.99)
P value for trend				0.364	<0.001	<0.001	<0.001
Age ≥40 y							
Q1 (<−12.84)	25 050	1444	57.6	0.90 (0.84–0.96)	1.17 (1.09–1.26)	1.11 (1.04–1.19)	1.01 (0.96–1.07)
Q2 (−12.84 to −4.12)	33 605	2061	61.3	0.98 (0.92–1.04)	1.07 (1.00–1.13)	1.06 (1.00–1.13)	1.02 (0.97–1.07)
Q3 (−4.11 to 2.26)	38 158	2358	61.8	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Q4 (2.27 to 10.53)	37 690	2231	59.2	0.96 (0.91–1.02)	0.96 (0.91–1.02)	0.96 (0.90–1.01)	0.92 (0.87–0.97)
Q5 (≥10.54)	31 625	1824	57.7	0.90 (0.84–0.95)	0.95 (0.89–1.01)	0.95 (0.90–1.01)	0.95 (0.90–1.01)
P value for trend				0.499	<0.001	0.001	<0.001

P=0.626 for the overall interaction between age and SDNN change for hypertension (multivariable-adjusted Model 1). HR indicates hazard ratio; PY, person-years; and SDNN, SD of normal-to-normal R-R intervals.

*Estimated from Cox proportional hazards models. Model 1 was adjusted for age, sex, center, year of screening examination, alcohol consumption, smoking, physical activity, body mass index, total energy intake, sleep duration, education level, depression, medication for hyperlipidemia, history of diabetes, and SDNN. Model 2 was adjusted for variables included in Model 1 and heart rate.

†Estimated from Cox proportional hazard models with quintiles of SDNN change, smoking, alcohol consumption, physical activity, total energy intake, body mass index, sleep duration, depression, medication for hyperlipidemia, and history of diabetes as time-dependent categorical variables, and age, sex, center, year of screening examination, education level, and SDNN at baseline as time-fixed variables.

autonomic neuropathy may be related to both HRV and hypertension. However, only 2% of our participants had diabetes (0.8% of young adults and 6.6% of older adults), and after adjusting for glucose level and history of diabetes, the results were similar. Thus, the overall findings of our study were likely unaffected by the lack of data on diabetic autonomic neuropathy. Finally, our study population consisted of relatively highly educated young and middle-aged Korean adults. Thus, these findings may not be generalizable to older individuals, other races and ethnicities, or populations with different demographics. However, our findings are less likely to be affected by survivor bias or bias related to unmeasured comorbidities compared with the findings derived from older populations. Moreover, while the rate of hypertension is increasing in young adults, the distinctive risk factors for this disease are not well established. Therefore, our findings suggest the importance of autonomic dysfunction measured using 10-second HRV as an independent and modifiable risk factor for hypertension in younger age groups.

Conclusions

Individuals, including young adults, with reduced 10-second HRV suggestive of autonomic nervous system abnormalities had a higher risk of developing hypertension. Furthermore, a protective association regarding the development of hypertension was observed in the group in which HRV increased over time. Even ultrashort-term HRV based on 10-second electrocardiography, which is widely used in clinical practice, is expected to play a role in stratifying and monitoring the risk of incident hypertension.

Perspectives

Among young (aged <40 years) and older adults, the 10-second HRV derived from routine standard electrocardiography and its change over time based on repeated measurements had a potential role as an independent risk factor for incident hypertension. This study suggests that autonomic dysfunction may contribute to the development of hypertension, with a slightly stronger effect in young adults. Owing to its wide applicability and low cost, there is a potential benefit for hypertension prevention and monitoring in clinical settings. However, further research is required to evaluate its utility.

ARTICLE INFORMATION

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Y. Chang and S. Ryu planned, designed, and directed the study, including quality assurance and control. S. Ryu analyzed the data and designed the analytical strategy of the study. Y. Chang, H. Shin, and S. Ryu supervised the field activities. All authors conducted a literature review and prepared the Materials and Methods and Discussion sections of the article. J. Kang, Y. Kim, and Y. Chang drafted the article. All authors interpreted the results and contributed to critical revision of the article. All authors have read and approved the final article. We thank the staff members of the Kangbuk Samsung Health Study for their hard work, dedication, and continued support.

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Disclosures

None.

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